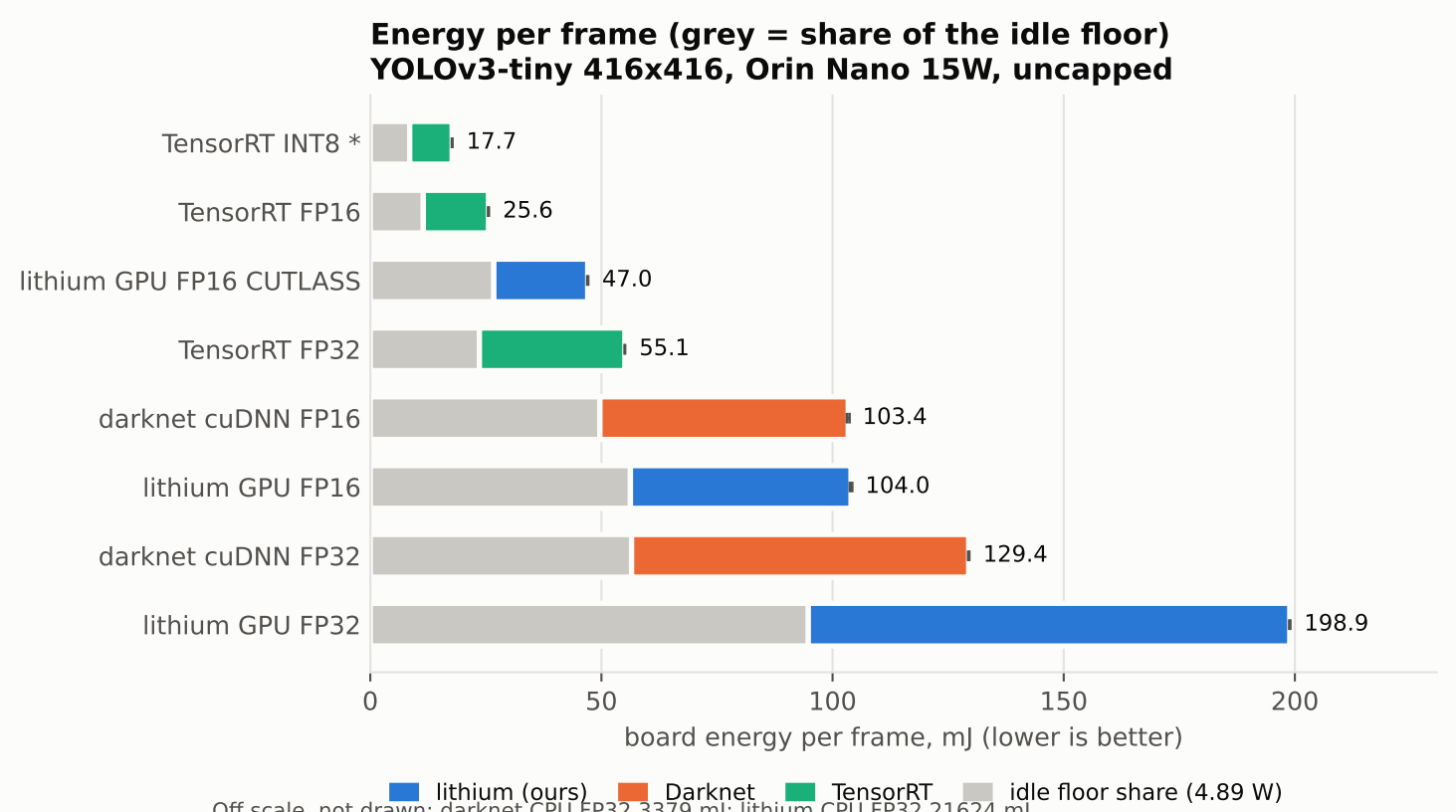
Energy per frame (grey = share of the idle floor)
YOLOv3-tiny 416x416, Orin Nano 15W, uncapped
TensorRT INT8 *
TensorRT FP16
lithium GPU FP16 CUTLASS
TensorRT FP32
darknet cuDNN FP16
lithium GPU FP16
darknet cuDNN FP32
lithium GPU FP32
17.7
25.6
47.0
55.1
103.4
104.0
129.4
198.9
0
50
100
150
200
board energy per frame, mJ (lower is better)
lithium (ours)
Darknet
TensorRT
idle floor share (4.89 W)
Off scale, not drawn: darknet CPU FP32 3379 mJ; lithium CPU FP32 21624 mJ
TensorRT INT8 *: built with trtexec --int8 and no calibration data; speed only